Himanshu Kumar

himanshuramgariya@gmail.com | +91-6207296193 | LinkedIn | Github

PROFILE SUMMARY

M.Tech Computer Science and Engineering student at **NIT Silchar** building **production-style Generative AI and Machine Learning applications**. Hands-on with **LLMs, RAG, NLP, vector databases, and Python backends**. Currently an **Embedded Linux Software Intern at ABB**. Seeking **AI/ML Engineer and Software Engineer roles**.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL, Bash

Generative AI and LLMs: LLMs, RAG, Prompt Engineering, AI Agents, LangChain, LangGraph, Hugging Face, Ollama, LoRA/QLoRA, MCP

Machine Learning and DL: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, NLP, Computer Vision

Vector Databases: FAISS, Pinecone, ChromaDB, PostgreSQL, MySQL, Firebase

Backend and Web Development: FastAPI, Flask, Django, Node.js, React.js, REST APIs, Streamlit, Tailwind CSS

MLOps, Cloud and Developer Tools: Docker, Git, MLflow, AWS, Linux/Unix, Postman, VS Code

Embedded and Core CS: Embedded Linux, systemd, Kernel Logs, Debugging, DSA, OOP, DBMS, OS, Computer Networks

EXPERIENCE

Embedded Systems (Linux) Software Intern, ABB – Bengaluru, India — Aug 2026 – Present

- Develop and debug embedded Linux apps in C/C++ and Python for ABB's ZEE600C substation automation platform.
- Diagnose embedded Linux issues by analysing boot, kernel, systemd, and runtime logs across device interfaces.

Open5G Intern, C-DAC – Pune, India — Jun 2024 – Aug 2024

- Deployed and configured Open5GS to implement 5G mobile core network functions.
- Simulated 5G networks with UERANSIM to analyse protocol behaviour between UE and the core network.
- Managed network deployments on Linux/Unix, strengthening system administration and shell scripting skills.

PROJECTS

MediRAG – AI-Powered Medical Question-Answering Assistant | **Live Demo**

Python, Flask, LangChain, Mistral AI, Pinecone, Hugging Face, BM25, SQLite, Docker

- Built a RAG-based medical chatbot using LangChain, Pinecone, and Mistral AI over 13,828 passages; hybrid search and reranking improved Hit@5 from 56% to 100%.
- Implemented safety guardrails and 10-language support with emergency alerts (112) in English and Hindi.
- Developed a full-stack Flask app with streaming responses, lab-report and image analysis, and voice input; Dockerized with GitHub Actions CI/CD (103 pytest tests), cutting memory from 2 GB to 240 MB, deployed on Render.

ShadowTrace – AI-Powered WhatsApp Chat Intelligence Platform | **Live Demo**

Python, Streamlit, NLP, LLMs, RAG, FAISS

- Built an AI-driven analytics platform processing 1M+ chat messages with 6 NLP modules (sentiment, emotion, topic, keyword, toxicity, language detection).
- Developed a RAG pipeline with FAISS vector search for context-grounded LLM question answering.
- Integrated 5+ LLM features: summarisation, highlights, recommendations, and conversational insights.

APEX – AI-Powered Fitness Tracker | **Live Demo**

Next.js 16, React 19, TypeScript, Supabase, PostgreSQL, Mistral AI, Gemini API, Three.js, Vercel

- Built a full-stack AI fitness platform with a Three.js 3D landing page and a PostgreSQL database with 6 tables.
- Developed an AI gateway (Mistral AI, Gemini) with fallback and usage limits for meal-photo analysis in under 3 seconds.
- Implemented Supabase auth and row-level security on all 6 tables, verified by 30+ access-control checks.
- Set up GitHub Actions CI/CD with 113 Vitest tests; 5 fitness-tracking modules serve 6 active users.

EDUCATION

National Institute of Technology Silchar, Assam – M.Tech, CSE | **CGPA: 7.94** — August 2025 – May 2027

Tezpur University, Assam – B.Tech, CSE | **CGPA: 8.01** — Oct 2021 – May 2025

Delhi Model Public School – 12th | **89%** — June 2018 – March 2020

ACHIEVEMENTS AND LEADERSHIP

- Solved 200+ Data Structures and Algorithms problems on LeetCode, GeeksforGeeks, and Coding Ninjas.
- Department Representative, NSS; Assistant Prefect, Patkai Men's Hostel, Tezpur University.